

Classical AI vs. LLMs for Decision-Maker Alignment in Health Insurance Choices

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Abstract

As algorithmic decision-makers are increasingly applied to high-stakes domains, AI alignment research has evolved from a focus on universal value alignment to context-specific approaches that account for decision-maker attributes. Prior work on Decision-Maker Alignment (DMA) has explored two primary strategies: (1) classical AI methods integrating case-based reasoning, Bayesian reasoning, and naturalistic decision-making, and (2) large language model (LLM)-based methods leveraging prompt engineering. While both approaches have shown promise in limited domains such as medical triage, their generalizability to novel contexts remains underexplored. In this work, we implement a prior classical AI model and develop an LLM-based algorithmic decision-maker evaluated using a large reasoning model (GPT-5) and a non-reasoning model (GPT-4) with weighted self-consistency under a zero-shot prompting framework, as proposed in recent literature. We evaluate both approaches on a health insurance decision-making dataset annotated for three target decision-makers with varying levels of risk tolerance (0.0, 0.5, 1.0). In the experiments reported herein, classical AI and LLM-based models achieved comparable alignment with attribute-based targets, with classical AI exhibiting slightly better alignment for a moderate risk profile. The dataset and open-source implementation are publicly available at: <https://github.com/TeX-Base/ClassicalAIvsLLMs> and <https://github.com/Parallax-Advanced-Research/ITM>.

1. Introduction

The advancement of large language models (LLMs) represents a new approach to building algorithmic decision-makers (DMs) in high-stakes domains (Liu et al., 2024; Hu et al., 2024; Guan et al., 2025). Unlike classical AI systems that rely on explicit rules and optimization procedures, LLM-based DMs draw on contextual reasoning and linguistic inference to approximate human judgment (Ravichandran et al., 2025; Chen et al., 2025b), offering an alternative for algorithmic decision-making. This raises a critical question: what advantages and limitations arise from each approach, particularly in their capacity to align with human decision-making attributes? In this work, we investigate how classical AI and LLM-based DMs can be designed to achieve decision-maker alignment (DMA), which occurs when algorithmic outputs reflect the reasoning processes of human decision-makers who operate under uncertainty, pressure, and limited resources (Sen et al., 2025). Under such conditions, where there is often no single correct answer, humans rely on cognitive attributes such as risk tolerance, cognitive reflection, and biases to guide their choices (Mainali & Weber, 2025). Designing algorithmic DMs that account for this variability requires methods that align not only with contextual factors but also with the diverse cognitive profiles that characterize human decision-making (Christian, 2020).

Recent alignment research has largely focused on enforcing fixed ethical principles, such as honesty, helpfulness, and harmlessness, through reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022; Bai et al., 2022). However, this approach does not necessarily translate to improved decision-making in dynamic, high-stakes contexts, as its rigidity limits adaptability to evolving, context-dependent scenarios (Fox, 2024). This limitation becomes particularly salient when values diverge across individuals, influenced by situational context, personal preferences, and social trade-offs. In such cases, algorithmic decision-makers must integrate moral commonsense reasoning and mediate conflicting preferences, which extend beyond the scope of universal alignment alone (Jiang et al., 2022; Sorensen et al., 2024). Designing algorithmic DMs that are alignable with these nuanced attributes remains a key challenge, demanding context-specific alignment that reflects how humans actually reason and decide, rather than how they are expected to choose optimally. Building on the notion of context-specific alignment, Hu et al. (2024) demonstrated that LLMs could be leveraged to align algorithmic decisions with cognitive attributes through zero-shot prompting and weighted self-consistency, potentially capturing nuanced human reasoning in dynamic, high-stakes contexts.

Classical AI approaches to decision-maker alignment employ structured reasoning techniques to emulate human decision-making under uncertainty, limited resources, and time pressure (Molineaux et al., 2024). Methods such as case-based reasoning, Bayesian inference, and naturalistic decision-making are employed to model how decision-makers utilize cognitive attributes when navigating complex choices (Molineaux et al., 2024). For example, Molineaux et al. (2024) proposed a case-based reasoning framework for combat medical triage, aligning the system outputs with decision-maker attributes using labeled prior cases. In this work, we extend a classical AI-based algorithmic decision-maker (DM) to a health insurance domain and implement an LLM-based algorithmic DM using Hu et al.’s methodology, adapting zero-shot prompting with weighted self-consistency on the same health insurance dataset and decision-maker targets defined in our experiment. This setup

allows a direct comparison of how each paradigm aligns with human cognitive attributes under consistent experimental conditions.

Although classical AI approaches to decision-maker alignment have been evaluated on the health insurance domain (Sen et al., 2025), LLM-based methods such as Hu et al. (2024) have not. Moreover, Hu et al.’s implementation poses practical challenges for direct evaluation in our context, including reliance on older models (LLaMA 2) with smaller context windows and lower reasoning performance compared to newer models, the need to download large model weights, and integration within a federated system that depends on external servers. These limitations make it difficult to apply the original implementation to our dataset and decision-maker targets. Our goal is to fill this gap by re-implementing the LLM-based methodology and evaluating both classical AI and LLM-based DMs on the same health insurance dataset with the same decision-making attribute, *risk tolerance*, designed specifically for our experiments.

For the LLM-based algorithmic DM, we develop a decision-maker using Hu et al.’s methodology to ensure reproducibility and compatibility with our experimental setup. We evaluate two models: GPT-5, a reasoning model, and GPT-4, a non-reasoning model, both using weighted self-consistency with zero-shot prompting. For the classical AI algorithmic DM, we re-implement the algorithm proposed by Molineaux et al. (2024), and evaluate both DMs on the same target decision-makers. This design enables a direct comparison of alignment accuracy across individuals with varying levels of risk tolerance. Our contributions are twofold: (1) we introduce a LLM-based DM tailored to risk-sensitive decision-making, fully self-contained without reliance on federated systems or external servers, and (2) we provide a systematic comparative analysis of classical AI and LLM-based DMs, offering insights into the design of adaptive, cognitively grounded algorithmic decision-makers.

The remainder of this paper is organized as follows. Section 2 reviews related work on algorithmic decision-makers and alignment strategies. Section 3 details our methodology, including the construction of the classical AI baseline, the LLM-based decision-maker, and the evaluation metric applied to the health insurance dataset. Section 4 reports results, highlighting the comparative strengths and trade-offs of each approach. Section 5 concludes with our key findings and the directions for future research.

2. Background

In this section, we first review research on decision-maker alignment, highlighting foundational work in value alignment and recent developments in pluralistic alignment. We then discuss both classical approaches and emerging methods that leverage LLMs to capture user- and domain-specific variability. Finally, we provide an overview of the dataset used in our experimental study.

2.1 Decision-Maker Alignment

Human decision-making is guided by internal mental models that shape how individuals perceive alternatives, evaluate risks, and select courses of action (Chermack, 2003). A central finding across cognitive science and psychology is that choices often diverge from the assumptions of rational choice theory, particularly in uncertain or high-stakes environments (Tversky & Kahneman, 1974).

In such contexts, decision makers are frequently forced to select suboptimal options due to environmental or cognitive limitations (Simon, 1955).

Decision-Maker Alignment (DMA) refers to the design of algorithms that align with the values and decision-making tendencies of the individual decision-makers. In scenarios where optimal decisions are unavailable, DMA focuses on modeling the variability in human choices by ensuring that the model’s decisions align with the attributes that guide each decision-maker (Molineaux et al., 2024; Sen et al., 2025). By explicitly considering different levels of decision-maker attributes, DMA provides a framework for creating AI systems that can adapt to differing perspectives and reasoning patterns.

2.2 Value Alignment vs. Pluralistic Alignment

Value alignment approaches aim to align algorithmic decision-makers with a fixed set of normative values. Standard methods often employ reinforcement learning from human feedback (RLHF), where a reward model is trained on human preference data to shape outputs in accordance with broad values such as honesty, harmlessness, and helpfulness (Ouyang et al., 2022). While effective at providing coarse reward signals, these methods are limited to static notions of human intent. To support broader alignability, benchmark datasets like ETHICS enable algorithmic DMs to align with values such as justice, well-being, virtues, commonsense morality, and duties (Hendrycks et al., 2021). Shen et al. (2025) further developed ValueCompass to evaluate alignment across real-world scenarios using 56 value statements from Schwartz’s Theory of Basic Values, highlighting the variability of values and limitations of static strategies (Shen et al., 2025).

These limitations have motivated more flexible alignment methods. Domain-aware alignment incorporates ethical principles along with domain-specific knowledge (Shetty et al., 2025; Rauch et al., 2025), while pluralistic alignment explicitly models cognitive and behavioral variation across users (Sorensen et al., 2024). Such approaches capture human reasoning variability by aligning AI systems with personalized mental models, user preferences, and decision styles (Chen et al., 2025a; Wu et al., 2025). Wu et al. (2025) proposed a dynamic method in which LLMs infer and adapt to user preferences through multi-turn interactions, while Guan et al. (2025) introduced a unified framework integrating preference memory, feedback-driven refinement, and individualized generation. Together, these works illustrate a shift from static value alignment toward pluralistic, context-dependent approaches that better reflect the diversity of human values and decision-making attributes, particularly for high-stakes algorithmic DMs.

2.3 Classical AI Approaches to Decision-Maker Alignment

While broader types of alignment approaches have been experimented with using expert knowledge, utility theory, and human-in-the-loop designs (Newell & Simon, 1956; Zhi-Xuan et al., 2024), they do not address the alignment we focus on here, which is user-specific, context-specific alignment with decision-making. To date, the only work exploring this practical and individual alignment using classical AI methods is Molineaux et al. (2024), who developed an algorithmic DM combining case-based reasoning, Monte Carlo simulation, Bayesian diagnosis, and naturalistic decision-making and demonstrated improved alignment in simulated military triage scenarios.